



Outlier Analysis - Outline

Introduction / Motivation / Definition

Statistical-based Detection

- Distribution-based, depth-based

Deviation-based Detection

- Sequential exception, OLAP data cube

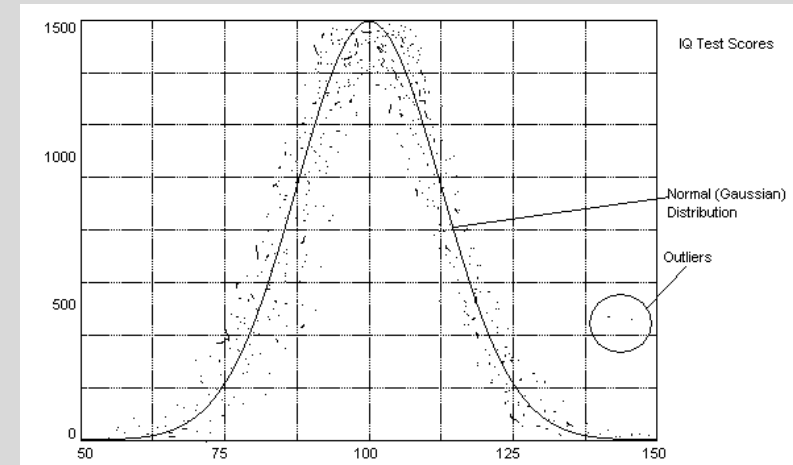
Distance-based Detection

- Index-based, nested-loop, cell-based, local-outliers

Questions

Statistical-Based Outlier Detection (Distribution-based)

- Assumptions:
 - Knowledge of data (distribution, mean, variance)
- Statistical discordancy test
 - Data is assumed to be part of a working hypothesis (working hypothesis)
 - Each data object in the dataset is compared to the working hypothesis and is either accepted in the working hypothesis or rejected as discordant into an alternative hypothesis (outliers)



Working Hypothesis: $H: o_i \in F$, where $i = 1, 2, \dots, n$.

Discordancy Test: is o_i in F within standard deviation = 15

Alternative Hypothesis:

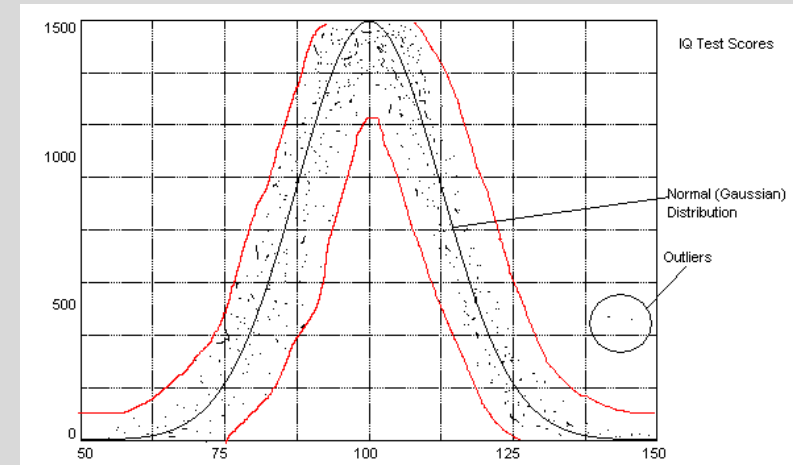
-Inherent Distribution: $\bar{H}: o_i \in G$, where $i = 1, 2, \dots, n$.

-Mixture Distribution: $\bar{H}: o_i \in (1 - \lambda)F + \lambda G$, where $i = 1, 2, \dots, n$.

-Slippage Distribution: $\bar{H}: o_i \in (1 - \lambda)F + \lambda F'$, where $i = 1, 2, \dots, n$.

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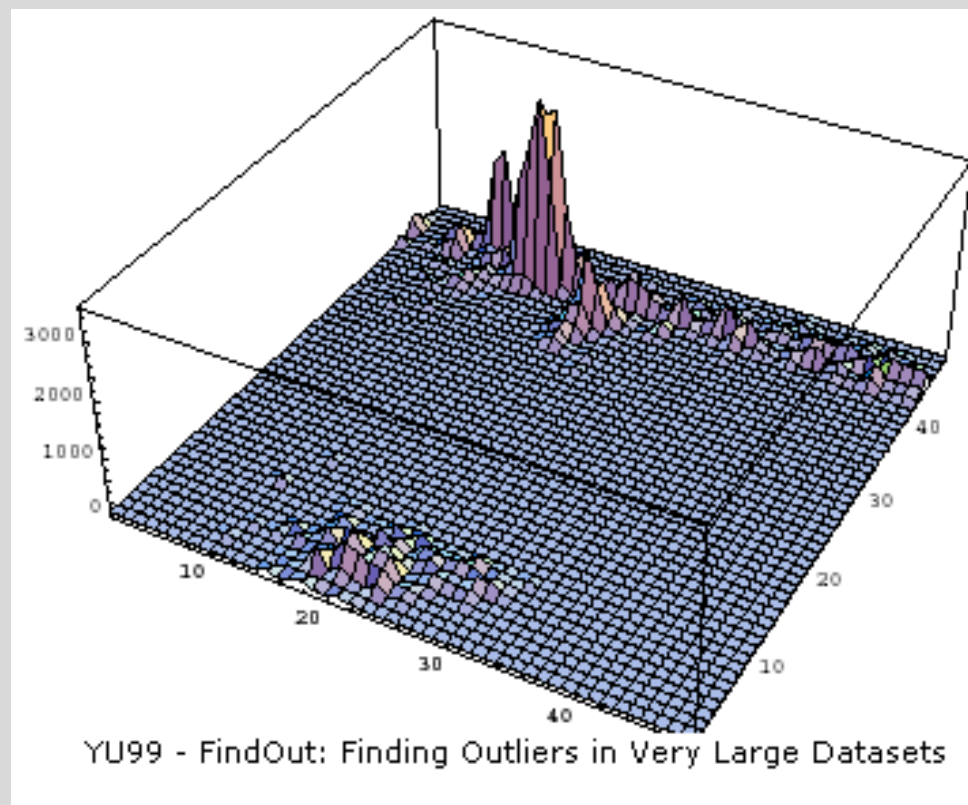
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Statistical-Based Outlier Detection (Depth-based)

- Data is organized into layers according to some definition of depth
- Shallow layers are more likely to contain outliers than deep layers
- Can efficiently handle computation for $k < 4$



Statistical-Based Outlier Detection

- Strengths
 - Most outlier research has been done in this area, many data distributions are known
- Weakness
 - Almost all statistical models are univariate (only handle one attribute) and those that are multivariate only efficiently handle $k < 4$
 - All models assume the distribution is known – this is not always the case
 - Outlier detection is completely subjective to the distribution used



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Deviation-Based Outlier Detection

Simulate a mechanism familiar to human being: after seeing a series of similar data, an element disturbing the series is considered an exception

Sequential Exception Techniques

OLAP Data Cube Techniques

Sequential Exception

- Select subsets of data I_j ($j = 1, 2, \dots, n$) from the dataset I
- Compare the dissimilarity of I and $(I - I_j)$ with $\sum_{i=1}^N (x_i - \bar{x})^2$
- Find out the minimum subset I_j that reduce the dissimilarity the most
- Smoothing factor $SF(I_j) := C(I - I_j) * (D(I) - D(I - I_j))$
 - D is a dissimilarity function
 - C is a cardinality function, for example, the number of elements in the dataset $\frac{1}{N} X$

Example

Let the data set I be the set of integer values $\{1, 4, 4, 4\}$

I_j	$I - I_j$	$C(I - I_j)$	$D(I - I_j)$	$SF(I_j)$
$\{\}$	$\{1, 4, 4, 4\}$	4	1.69	0.00
$\{4\}$	$\{1, 4, 4\}$	3	2.00	-0.93
$\{4, 4\}$	$\{1, 4\}$	2	2.25	-1.12
$\{4, 4, 4\}$	$\{1\}$	1	0.00	1.69
$\{1\}$	$\{4, 4, 4\}$	3	0.00	5.07
$\{1, 4\}$	$\{4, 4\}$	2	0.00	3.38
$\{1, 4, 4\}$	$\{4\}$	1	0.00	1.69

Note, when $I_j = \{\}$, $D(I) = D(I - I_j) = 1.69$, $SF(I_j) = 0$

When $I_j = \{1\}$, $SF(I_j)$ has the maximum value, so $\{1\}$ is the outlier set



OLAP Data Cube Technique

- Deviation detection process is overlapped with cube computation
- Precomputed measures indicating data exceptions are needed
- A cell value is considered an exception if it is significantly different from the expected value, based on a statistical model
- Use visual cues such as background color to reflect the degree of exception